Weixuan Yuan

Zürich, Switzerland · weixuan.yuan@inf.ethz.ch · yzll0.github.io

Research internship interests: LLM reasoning and post-training; AI for mathematics and theoretical computer science; graph algorithms.

Education

ETH Zürich — Ph.D. in Computer Science May 2025–present
Advisor: Prof. Rasmus Kyng. Research in theoretical computer science and algorithms, with AI-related research.

ETH Zürich — M.Sc. in Data Science September 2022–October 2024
GPA: 5.92/6.00.

Xi'an Jiaotong University — B.Sc. in Mathematics September 2018–July 2022
GPA: 93.43/100.

Research Experience

ETH Zürich — Research Assistant, Rasmus Kyng's group October 2024–April 2025
Research on deterministic graph algorithms; continued as a Ph.D. student from May 2025.

AuXSpec: executable specifications and LLM reasoning

Co-first author; under review at ICLR 2027.

- Proposed the executable-postcondition approach and contributed substantially to the Lean specification format, linking declarative specifications to executable counterparts through equivalence theorems.
- Co-developed a multistage, LLM-based agentic data synthesis pipeline using executable checks as verifiable feedback.
- Independently completed expert review of 300 benchmark problems and audited 60 generated outputs, confirming 55 as correct.
- Contributed to discussions of post-training methods, reward design, results, and analysis with the co-first author responsible for training.
- **Team results:** 3,037 LeetCode problems with proven equivalence and 19,492 SFT samples. After SFT/RL, Qwen3.8-27B improved SpecGen pass@4 from 0.7% to 60.3% on AuXSpec-Test and from 21.9% to 84.7% on the Verina out-of-distribution benchmark.

Deterministic graph algorithms

- Co-authored the first deterministic almost-linear-time algorithm for Gomory-Hu trees, with running time $m^{1+o(1)}$ (*FOCS 2025*).
- Presented the paper at FOCS 2025. [slides]

Selected Publications

Deterministic Almost-Linear-Time Gomory-Hu Trees
Amir Abboud, Rasmus Kyng, Jason Li, Debmalya Panigrahi, Maximilian Probst Gutenberg, Thatchaphol Saranurak, **Weixuan Yuan**, Wuwei Yuan.
FOCS 2025. [paper]

Optimal subsampling for principal component analysis
Xuehu Zhu[†], **Weixuan Yuan**[†], Zongben Xu, Wenlin Dai.
Science China Mathematics 68, 2993-3016, 2025. [paper]

Optimal subsampling for estimation of dimension reduction directions
Xinru Jia[†], **Weixuan Yuan**[†], Xingqiu Zhao, Xuehu Zhu.
Scandinavian Journal of Statistics 53(1), 575-611, 2026. [paper]

[†] Equal contribution. Author order in theoretical computer science papers is alphabetical. The statistics research was carried out during undergraduate studies (before 2023).

Under-Review Manuscript

AuXSpec: Learning to Autoformalize Program Specifications via Executable Postconditions
Co-first author; under review at ICLR 2027. Not yet publicly available.

Honors

Willi Studer Prize, ETH Zürich 2025
Best graduate of the Master's programme in Data Science. [announcement]

Technical Skills

Tools: Python; Lean 4 and formal verification.
Research background: Graph algorithms; mathematical and statistical methods; optimization, including gradient-based optimization.

Teaching

Teaching Assistant at ETH Zürich for *Big Data* and *Algorithms & Probability*.